MAR-ACC-49 Arista Access Node QoS Congestion Event Resolution

Node: MAR-ACC-49 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_5047

Description

This Standard Operating Procedure (SOP) outlines the steps required to diagnose and resolve QoS congestion events, specifically microbursts, interface tail drops, and QoS queue starvation, on the MAR-ACC-49 node, which is an Arista Access node located at the Marathahalli Core site. The event is triggered by the alarm code ALARM_5047, indicating a major severity failure. The underlying protocol mechanics contributing to this failure domain are the interactions between the QoS policies, class maps, and queueing mechanisms in the hardware. The congestion event is caused by the excessive creation of microbursts, which are bursts of packets that exceed the buffer capacity of the interface, leading to tail drops and subsequent QoS queue starvation. This can occur when the QoS policies are not properly configured or when there are configuration inconsistencies, resulting in uneven traffic distribution across the queues. The hardware mechanics involved include the buffer memory, scheduling algorithm, and queue depth, which all play a critical role in determining the packet drop behavior. Understanding these mechanisms is crucial in diagnosing and resolving the QoS congestion events.

Pre-requisite Checks

- Ensure remote hands are dispatched to the Marathahalli Core site to perform physical checks and identify any environmental factors that may be contributing to the failure.

- Verify the status of the redundant link(s) to ensure they are functioning correctly and not contributing to the congestion event.

- Check the power supply to the node to ensure it is functioning within specifications.

Potential Root Causes

Cause	Probability
Misconfigured QoS policies leading to uneven traffic distribution across queues	High
Configuration inconsistencies resulting in excessive microburst creation	Medium
Hardware buffer memory issues	Low

Diagnostic Commands & Expected Outputs

```
Command: show qos policy-map, Expected Output: Policy Map:
mar-acc-49-policy-map Class Map: mar-acc-49-class-map Queue:
mar-acc-49-queue Buffer: mar-acc-49-buffer Drop Threshold: 100 Queue
Depth: 1000, Description: This command displays the QoS policy map,
class map, and queue information, allowing us to verify the
configuration and identify any inconsistencies.
```

```
Command: show interface statistics, Expected Output: Interface:
mar-acc-49-interface Packets Transmitted: 1000000 Packets Received:
1000000 Errors: 100 Drops: 1000, Description: This command displays
the interface statistics, allowing us to verify the packet count,
errors, and drops, which can help identify the source of the
congestion event.

Command: show queue statistics, Expected Output: Queue:
mar-acc-49-queue Packets in Queue: 10000 Drops: 1000 Queue Depth:
1000, Description: This command displays the queue statistics,
allowing us to verify the packet count, drops, and queue depth, which
can help identify the source of the congestion event.
```

Resolution Steps

Step 1: Step: Configure QoS policies to ensure even traffic distribution across queues, Command: configure terminal policy-map mar-acc-49-policy-map class mar-acc-49-class-map queue mar-acc-49-queue buffer mar-acc-49-buffer drop-threshold 50 queue-depth 1000, Description: This step involves configuring the QoS policies to ensure even traffic distribution across queues, which can help prevent microburst creation and subsequent congestion events. The buffer drop threshold and queue depth are adjusted to prevent excessive packet drops.

Step 2: Step: Verify QoS policies and configuration, Command: show qos policy-map, Description: This step involves verifying the QoS policies and configuration to ensure that the changes made in the previous step are correct and effective.

Step 3: Step: Verify interface and queue statistics, Command: show interface statistics show queue statistics, Description: This step involves verifying the interface and queue statistics to ensure that the congestion event has been resolved and the packet drop behavior has improved.

Post-Check Verification

```
Command: show qos policy-map, Expected Output: Policy Map:
mar-acc-49-policy-map Class Map: mar-acc-49-class-map Queue:
mar-acc-49-queue Buffer: mar-acc-49-buffer Drop Threshold: 50 Queue
Depth: 1000, Description: This command is used to verify that the QoS
policies and configuration have been successfully applied.

Command: show interface statistics, Expected Output: Interface:
mar-acc-49-interface Packets Transmitted: 1000000 Packets Received:
1000000 Errors: 0 Drops: 0, Description: This command is used to
verify that the packet drop behavior has improved and the interface
statistics are within normal range.

Command: show queue statistics, Expected Output: Queue:
mar-acc-49-queue Packets in Queue: 0 Drops: 0 Queue Depth: 1000,
Description: This command is used to verify that the queue statistics
are within normal range and the packet drop behavior has improved.
```

Roll-back Procedures

- Step: Restore original QoS policies and configuration, Command: configure terminal rollback mar-acc-49-policy-map , Description: This step involves restoring the original QoS policies and configuration to prevent any disruption to the network.

- Step: Verify QoS policies and configuration, Command: show qos policy-map, Description: This step involves verifying the QoS policies and configuration to ensure that the changes have been successfully rolled back.